



Supporting Online Material for

Trace Metals as Biomarkers for Eumelanin Pigment in the Fossil Record

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Published 30 June 2011 on *Science Express*

DOI: 10.1126/science.1205748

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Materials and Methods

Materials

1. *Confuciusornis sanctus* (MGSF315) is on loan to the University of Manchester from the Chinese Institute for Vertebrate Paleontology and Paleoanthropology (IVPP). See Figure 1 for photograph. (MGSF accession prefixes refer to samples held in the collection of the University of Manchester School of Earth, Atmospheric, and Environmental Sciences).

2. The second *Confuciusornis sanctus* (LL12418) is on loan to the Manchester Museum from the IVPP; photograph below (scale bar 10 cm total, markings 1 cm each). (LL and BB accession prefixes refer to the Manchester Museum, a public museum affiliated with the University.)



3. BHI-6358 is a small exceptionally preserved feather (unidentified) approximately 2 cm in length (for photograph see Figure 3A). Accession numbers with a BHI prefix refer to the Black Hills Institute Museum, part of the Black Hills Institute (Hill City, South Dakota).

4. BHI-6319 is a single slab from the Green River Formation (26) consists of a small slab with a preserved fish (*Gosiutichthys parvus*, ~2.5 cm in length) and feather (unidentified, ~2cm in length). For photograph see Figure 3B.

5. *Gansus yumenensis* (MGSF317) is a single feather collected by one of the authors (H. You) from the Lower Cretaceous (~115 to 105 Mya) Xiagou Formation near Changma, Gansu Province northwestern China (3). Phylogenetic analysis places *Gansus* within the Ornithurae, making it the oldest known member of the clade (3). The anatomy of *Gansus*, like that of other

non-neornitheat (nonmodern) ornithuran birds, indicates specialization for an amphibious life-style (3).

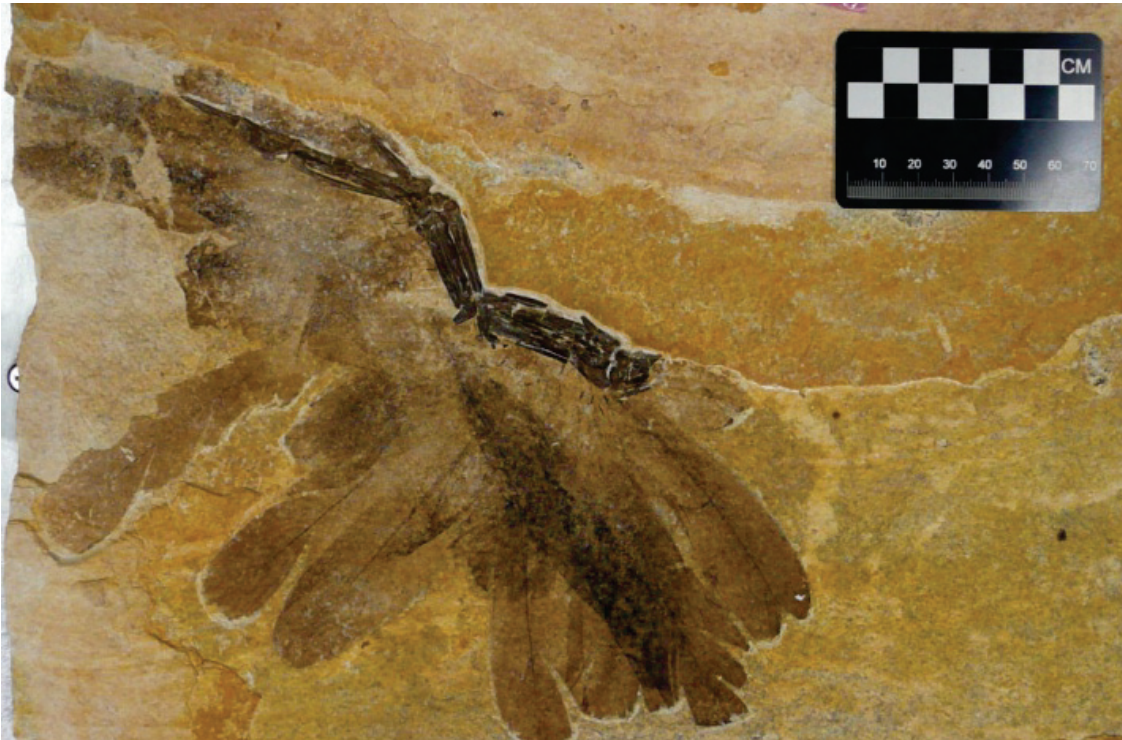
6. A single black feather identified tentatively as from *Haliaeetus leucocephalus* (Figure 3D) was generously provided from a captive bird at a bird sanctuary.

7. Blue Jay feather (Figure 3E; *Cyanocitta cristata*) was provided by the Manchester Museum (specimen BB.9013.1).

8. Fossil squid *Rachiteuthis* tent. (BHI-2243B) is from Hakel, Lebanon (Late Cretaceous, ~90 Mya). See Figure 3F for photograph.

9. Extant squid (Figure 3G; *Sepia officinalis*) was purchased from a fisherman's market.

10. Green River Feathered Wing (unidentified; HMNS 2010.185.02) is curated by the Houston Museum of Natural History who graciously allowed us access. See photograph below for details (note scale bar at upper right).



11. Green River Feather BHI-6403 (unidentified) is pictured below. Note scale bar at bottom left.



12. *Archaeopteryx lithographica* holotype (MB.Av.100) pictured below was graciously loaned by the Museum für Naturkunde, Humboldt University, Berlin. Feather is approximately 7 cm long and 1.5 cm wide.



13. Natural melanin (*Sepia officinalis*) was obtained commercially from Sigma-Aldrich. FTIR was completed on an as received portion. In order to produce Cu-melanin complexes for EXAFS and XANES standards, a part of this natural melanin was prepared and reacted with a copper solution as summarized below. This protocol is based on previously reported methods (21, 23).

I. Remove the metals already bound to the melanin:

- 1) Mix 50 g melanin with EDTA (10ml)
- 2) Centrifuge mixture at 2200 rpm and retain the supernatant
- 3) Add nanopure (DI) water (10ml) to the pellet and mix
- 4) Centrifuge and remove the water
- 5) Repeat this wash 3 more times with nanopure water

II. Saturate the melanin with copper:

- 6) Add 50 ml of $\text{Cu}_2\text{SO}_4 \cdot 5\text{H}_2\text{O}$ 0.1M and leave overnight
- 7) Centrifuge mixture and keep the supernatant
- 8) Add approximately 10ml of HCl pH 4, stir, centrifuge and remove the supernatant
- 9) Repeat the HCl wash once more
- 10) Repeat the procedure twice more with nanopure (DI) water
- 11) Freeze dry

Methods

1. *SRS-XRF Imaging* was performed at wiggler beam lines 6-2 at SSRL. The beam line was operated in its standard configuration with a collimating mirror upstream of a Si (111) monochromator, and a pair of total reflection focusing mirrors downstream. For light element XRF images (Cl, S, P, and K) the excitation energy was chosen at 3.15 keV, for the heavier element images (Cu, Ca, Zn, Fe, Mn, Se, Ba, and Pb) an excitation energy of 12.0 or 13.5 keV was chosen. Flux at the sample surface varied between approximately 10^{10} and 10^{11} photons s^{-1} depending on the specific analytical conditions. At low incident energy light elements such as phosphorous and sulfur were easily resolved. Increasing the incident energy of the beam allowed analysis of heavier elements, especially the transition metals. The beam was focused onto a 100 μm thick tantalum pinhole of 50, 80, or 100 μm diameter and placed at a distance varying for the high energy measurements between approximately 1 and 8 mm from the fossil surfaces. Due to the very long focal depth of the setup, the beam striking the sample was only insignificantly larger than the pinhole diameter. Samples were mounted at either 45° (high incident energies) or 67° (low incident energy) relative to the incident beam. A photon counting single element silicon drift detector (Vortex, SII NanoTechnology USA Inc.) combined with Gaussian shaping amplifiers (Canberra) employing 0.125 μsec shaping times plus single channel analyzers was used to detect the XRF signals. For each element, the electronic windows were set to capture the fluorescent photons from the $K\alpha$ or $L\alpha$ emission lines. The width of the electronic windows corresponded to typically 200 – 350 eV per fluorescence line.

For scans at high incident beam energy the detector was placed at a 90° angle to the beam in order to minimize the unwanted scattering signal. For scans at 3.15 keV where the scattering was much smaller the detector was placed normal to the sample

surface. All fossil and extant samples were held within a purpose built sample chamber which was carefully mounted on a computerized x-y translational stage and rapid scans were performed by continuously translating the sample horizontally across the beam. At the end of each horizontal line a vertical step (50, 80, or 100 μm /step, depending on pinhole) was performed, and the horizontal scan direction was reversed. Data were recorded on the fly in both horizontal directions at a rate corresponding to a travel distance of 50, 80, or 100 μm (to match pinhole size) per ~ 3 ms readout. The beam intensity I_0 was monitored with a He filled ion chamber upstream of the pinhole. During scanning, small drifts of the beam with respect to the pinhole were corrected periodically.

For the light element XRF imaging an X-ray transparent window was placed on the sample chamber and the chamber was purged of air with helium. Window material was an ~ 30 μm thick polyethylene film that touched the detector and was located ~ 2 cm from the pinhole. Sample surface to detector distance was varied from 200 to 14.5 mm, depending on the major element fluorescence intensity. At high energy, in order to reduce the very large Ca signal we furthermore placed a 50.8 μm thick Al foil in front of the detector. Both the air absorption and the Al foil reduced the Ca $K\alpha$ fluorescence by a factor of 500 whereas the 3d transition metal signals were only reduced by between 20 and 80%.

Supplementary scans of BHI-6358 and BHI-6319 were completed at 11.5 keV incident beam energy on beam line 10-2a using a 100 μm pinhole and similar scanning geometry. In this case the detector was placed at 10 cm distance from the point of analysis and the fluorescence signal was recorded “on the fly” at 5 ms time intervals corresponding to a travel distance of 100 μm to match the pinhole spot size. Detector output was collected using digital xMAP x-ray spectrometers from XIA in buffered mapping mode. In order to synchronize the readout with the continuous stage motion, the signal was gated electronically using TTL output from the stage motion to advance each pixel.

SRS-XRF images presented in the main text and below represent raw data that have not been processed except for maximum intensity being clipped at between 90 and 99% of the maximum value in order to best show contrast relative to the background sedimentary matrix. For actual elemental concentration values, see point analysis information below.

2. *Micro-XRF* images were collected at the Stanford Synchrotron Radiation Lightsource (SSRL) using beam line 2–3. The incident x-ray energy was set to 10 keV using a Si (111) double crystal monochromator. The fluorescence lines of the elements of interest, as well as the intensity of the total scattered X-rays, were also monitored using a silicon drift Vortex detector. In addition to these regions of interest, the entire fluorescence spectrum was also collected at each data point. The microfocused beam of 2×2 μm was provided by a Rh-coated Kirkpatrick-Baez mirror pair (Xradia Inc.) The incident and transmitted x-ray intensities were measured with nitrogen-filled ion chambers. Samples were mounted at 45° to the incident x-ray beam and were spatially rastered in the microbeam using a Newport VP-25XA-XYZ stage. Beam exposure was 100 ms per pixel.

3. *XRF Point Analyses* were completed on beam line 6-2 by driving the rapid scanning stage to a location of interest defined by previously acquired maps, counting for 100 s (as compared to the ~3.3 ms per pixel used in rapid scanning), and acquiring a full EDS spectrum. While SRS-XRF intensity maps are excellent indicators of elemental distributions on the surfaces of fossils and other materials, without an understanding of topography, layer thickness, and layer composition they are semi-quantitative. In order to fully quantify elemental concentrations we estimate the fossilized layer major element composition and thickness, and measure the distance between sample surface and the x-ray detector. Then point analyses data are collected. We note that limits of detection are several atomic weight percent for low atomic weight elements, decreasing to approximately 1 ppm for heavier elements such as arsenic. In all cases point analyses compared well to yields at the same pixel for each element produced during scanning. Point analyses were processed using the PyMCA programme (27) using a full set of recorded geometrical parameters for each point analysed. Standardization was achieved using point analyses of bone apatite (Ca and P) or extant feathers (S) as an internal standard.

4. *XANES* were recorded in fluorescence mode on either beam line 6-2 or 2-3 by scanning the incident beam energy (via monochromator rotation) through the copper K edge and recording the emitted intensity of the copper K α line as a function of incident energy. Spectra were monitored during collection to avoid photoreduction of copper by the incident beam. If edge position shifted in successive scans, collection times were decreased and the incident beam was moved by several microns between successive scans in order to minimize photoreduction. A copper metal foil standard was used to calibrate the energy of the monochromator position. Linear Combination Analysis of Cu XANES spectra was used to fit unknown spectra by summing two standard spectra:

1) CuO, and 2) Cu-melanin. Fits were achieved using the Linear Combination Fitting routine for XANES data in ATHENA (28) where χ^2 minimization is used to find the best fit. Fitting was attempted two ways: 1) using the simple normalized absorption coefficient, and 2) using the first derivative of the absorption coefficient. The simple method was unsuccessful at reproducing the observed spectra. The derivative fitting method provided the most consistent results, and showed that for all fossil samples the copper near edge spectrum is dominated by a Cu-melanin coordination environment. We note however that the CuO component is significantly over-estimated by this fitting process, because we see no appreciable CuO determined via EXAFS for the *C. sanctus* specimen or Green River Feather (BHI-6358). Therefore the values below represent the absolute MINIMUM organic Cu-chelate present in these fossils. The method rejected attempts to include Cu metal into the fit as a third component, lending confidence to the results for the binary fitting reported below. Unsurprisingly, the fossilized squid matches pure melanin the best.

5. *EXAFS* were also recorded in fluorescence mode on beam line 2-3 at the Cu K edge. The energy range went from 200 eV below the edge (set to 8980 at the metallic Cu foil) to a *k* of approximately 12. Data were analyzed with Sixpack (29) fit with FEFF6 using reference scattering paths. Full details of the EXAFS fitting results are provided in the table and figures below. Number of atoms ($n \pm 1$), radial distance ($R \pm 0.01 \text{ \AA}$), and the

Debye-Waller factor (σ^2) are all best fits for all shells at the optimized local coordination geometry: i.e. the geometry corresponding to the minimum reduced χ^2 .

6. *VP-FEG-SEM* was completed using a Zeiss Supra40 instrument at the DRIAM Analytical Service, Dalton Research Institute, Manchester Metropolitan University. Accelerating voltages ranging from 1 kV to 15 kV were used at working distances from 16 mm to less than 3 mm. These ranges were used to optimise images for each sample. Both secondary electron (SE) and variable pressure secondary electron (VPSE) images were collected, depending on the nature of surface charging on the samples. For all images presented here absolutely no gold or carbon coating was used: all samples were analyzed without any chemical or physical alteration to their surface. Standard clean laboratory sampling handling protocols were used throughout. Point analyses of elements present were acquired using standardless energy dispersive spectroscopy (EDS, Oxford Instruments) at 20 kV and a working distance of 15 mm. Spectra collected were fitted and elements quantified using proprietary ZAF-corrected software. Counting times in excess of 100 live seconds were performed to maximise the signal-to-noise ratio of the spectra.

7. *FTIR* imaging was performed using a Perkin-Elmer Spotlight 400 with a 100 μm x 100 μm aperture in contact ATR mapping mode. Every point on these maps records a full range infra-red spectrum (4000 to 800 cm^{-1} with 4 cm^{-1} resolution). Maps are produced by selecting a specific absorption band and then displaying the intensity of absorption (or transmission) within each spectrum recorded at each point in the area mapped. All spectra and maps were background subtracted. Organic peak assignments were made using the Bio-Rad KnowItAll Informatics System 8.2 Multi-Technique database. Point FTIR spectra were acquired both from the Perkin-Elmer instrument and from a Bio-Rad Stingray FTS6000, also in ATR mode at 4 cm^{-1} wavenumber resolution.

8. *Image Analysis*: Image CorrelationJ (30) was used to calculate correlation coefficients and correlation maps. The algorithm uses regression analysis to compare pixel intensity values between two images either using single pixel values or based on a user defined local area size. Our correlations all used a local area size of 3 pixels. Output includes a global correlation coefficient for the entire image comparison plus an intensity correlation map. Comparison between SRS-XRF elemental maps required several steps. First, the SRS-XRF raw data was converted to a TIFF image clipped at the 99th percentile in order to give best image contrast. Then, the image was cropped to highlight the fossil area that included the region mapped using FTIR. We did not try to match contrast and brightness between SRS-XRF images. Images were cropped to carefully compare the same areas. Finally, the Image CorrelationJ plugin (31) was used to calculate a correlation coefficient and correlation map. The resulting greyscale correlation map was not processed in any way.

SOM Text

Geochemistry and Taphonomy

1. *Geochemistry*- Cu and Zn were analysed via SRS-XRF and given in ppmW, C was analysed by VP-FEG-SEM and given in wt%. As the original keratin breaks down, carbon may be lost and Cu concentrated into the remaining organic matrix. Keratin is originally 46.5% C, and so VP-FEG-SEM analysis allows us to estimate carbon loss and possible concentration effects on Cu. The range of copper concentrations that we observe in fossil specimens is 67 to 434 ppm. Our measured maximum trace metal concentration in a eumelanin pigmented extant feather is ~1,000 ppm Fe in a Blue Jay. Furthermore, we know that soft tissue is not stable after death and that there is likely to be mass loss during degradation. Using the measured carbon content of the organic residue in MGSF315 and if we assume both that copper has been conserved and that the original carbon content was equal to the known value in beta keratin of 46.5%, then we can correct our measured values for original concentration. Corrected in this way, the copper concentrations we measure in the fossils represent a range in the original feathers of 5 to 130 ppm Cu.

Copper (or Fe, Ni, Zn, Ca) added by geochemical processes might become melanin chelated, however the levels we see do not require significant additions from external sources. It is possible that Cu-chelation could drive Cu release from nearby minerals but this would leave a telltale Cu-depleted aureole around the fossil and we do not observe such features.

2. *Taphonomy*- Our results may help explain the process of pigment preservation. Initially, rapid burial is required in order to limit aggressive aerobic decay (32). In a relatively short period of time feathers may undergo a 90% mass loss (33). Originally, Cu, Fe, Ni, Zn, and Ca will be present as melanin chelates. At the onset of anaerobic degradation melanin is unsaturated with metal, and so as soft tissue breaks down and releases metals (Cu, Ni, Zn and Ca especially) they may partition into and add to the original organically bound metals within melanin. Copper and other metal zoning we see is thus inherited from the original melanin, plus a possible minor contribution from other organic chelates present in the soft tissue.

How might this metal biomarker distribution then survive over geological time? Recent work (34) examining soft tissue breakdown over extended periods (2 to 4 years) suggests that dehydration early in the taphonomic history of a specimen may shut down anaerobic breakdown processes. Copper itself inhibits bacterial degradation and archaeological evidence shows how bacteriostatic copper may preserve organic fibres for over 1,300 years (35). Furthermore, the presence of melanin itself also may play a key role in preservation with recent work suggesting that high melanic content might be an evolutionary response to the presence of feather-degrading bacteria (36).

Cu-bearing melanised tissue is therefore extremely resistant to anaerobic decay and, under the right conditions, may allow soft tissue to survive for extended periods until formation dewatering creates fully dehydrated conditions which preclude further microbial or solvent attack. For the *C. sanctus*, *G. yumenensis*, and Green River feathers discussed here, rehydration did not occur, at least not for long enough periods to destroy

or mineralize the specimen, and thus we can still resolve trace-metal distribution and organic functional groups most likely derived from the original melanin. This taphonomic pathway also leaves the structures of some melanosomes intact as required by other recent work (4, 5). For some samples, trace metal distributions may survive as a pigmentation indicator even after rehydration has destroyed all traces of structure and organic chemistry. Due to the fact that eumelanin is difficult to quantify even in fresh material, SRS-XRF may prove useful even in studying extant feathers.

Image analysis of *C. sanctus* (MGSF315) compositional maps was also completed to further constrain the chemical relationships between key elements (see SOM Figure S3 and Table S1). Correlation between copper and the other three elements presented here is positive and relatively high (R^2 : Cu/Ca=0.56, Cu/S=0.48, Cu/Zn=0.37) and the strongest relationships unquestionably occur within the feather regions.

In order to finally conclude whether trace metals indicate pigmentation or imagination, the salient observations are:

1) Ca, Zn, and Cu all map together in *C. sanctus*, consistent with biochemical chelation but inconsistent with geochemical precipitation.

2) X-ray spectroscopy rules out the presence of Cu oxides or sulfides as a significant host for copper, indicating that Cu is predominantly present as an organic copper chelate.

3) Electron microscope imaging analysis of small chips sampled from the MGSF315 (*C. sanctus*) show that areas of high copper in the neck and tail feathers contain possible fossilized eumelanosomes.

4) Infra-red spectroscopy indicates the absorption spectrum of chips taken from the neck of MGSF315 (*C. sanctus*) are directly comparable to the absorption spectrum of extant *Sepia officinalis* eumelanin (see SOM Fig. S13).

5) SRS-XRF maps and chemical analyses of feathers with completely different ages and geochemical conditions also show trace metal distribution patterns and chemistry controlled by the soft tissue.

6) X-ray and infra-red spectroscopy both produce results from other specimens, including the holotype of *Archaeopteryx lithographica*, that are consistent with the data from *C. sanctus* (see SOM Fig. S2).

7) XRF and SEM point analyses constrain the copper concentrations mapped and they do not require a massive influx of copper from external sources to produce the observed patterns (see SOM Tables S3 and S4 for further details).

8) Finally, we have completed measurements with non-avian species to provide a further test of our hypothesis that copper zonation in fossilized soft tissues represents original eumelanin pigmentation distribution. Copper distributions in fish eyes and squid ink sacs are easily resolved and correlate perfectly with tissue expected to have high melanin content.

SOM Fig. S9 presents a summary diagram explaining how residual melanin chemistry is preserved.

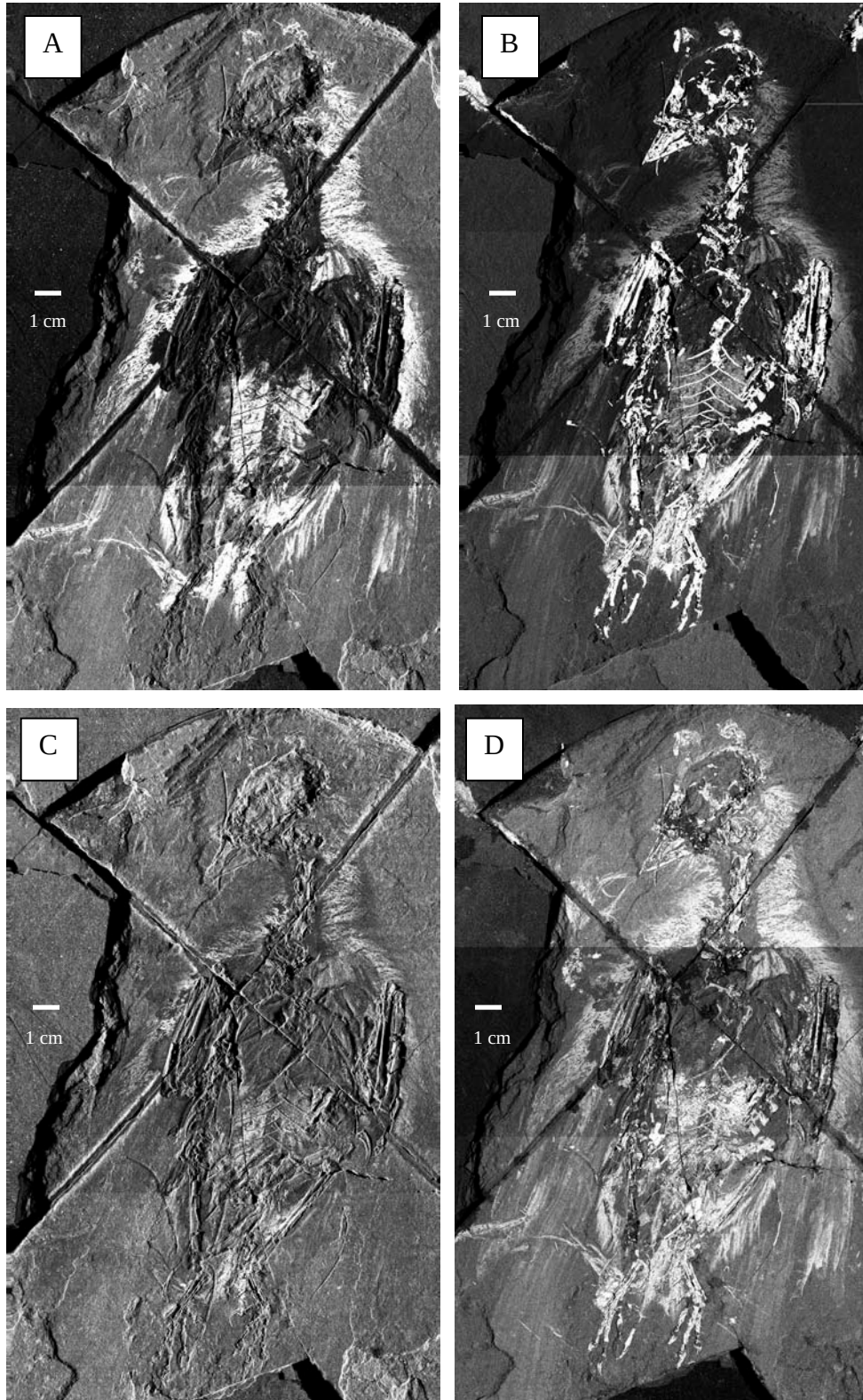


Fig. S1
 SRS-XRF images of *C. Sanctus* showing clear correlation of A) Cu, B) Ca, C) Zn, and D) S within the downy body feathers and portions of the flight feathers.

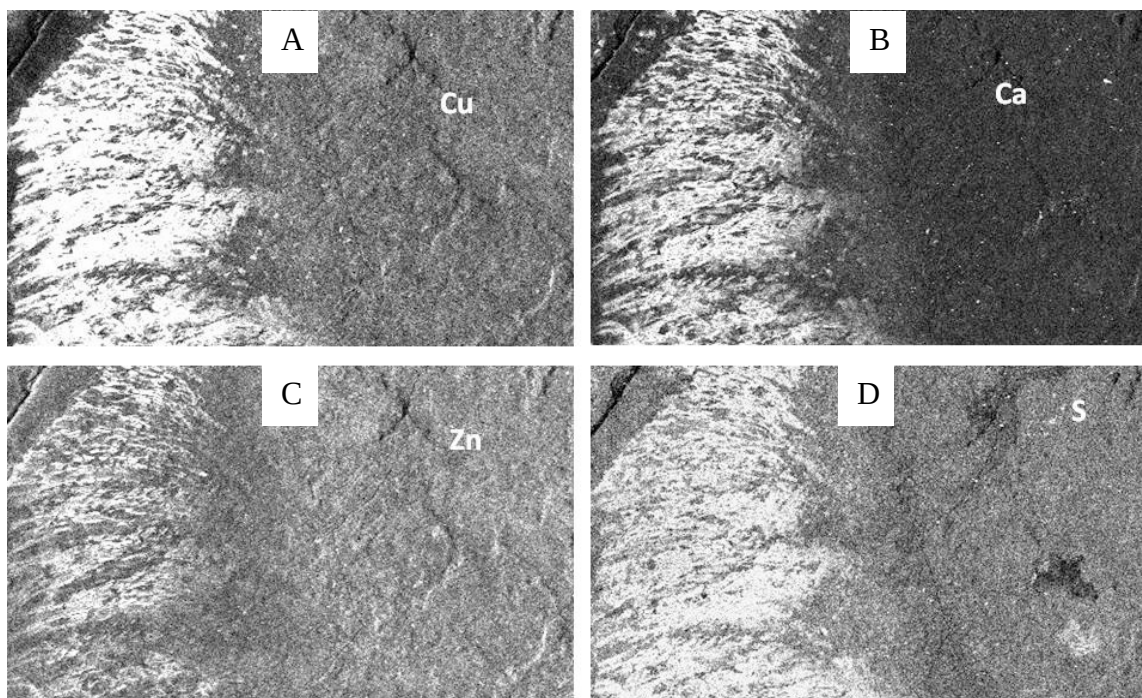


Fig. S2

Detail of A) Cu, B) Ca, C) Zn, and D) S within the downy neck region (area highlighted as yellow box in Figure 1A). The most obvious aspect is that when magnified the feather structures of these chemical zones becomes clear. It is also important to note that while the biochemistry of Cu, Ca, and Zn in feathers is strongly correlated and hence this distribution is predictable based on melanin chelation, the geochemical relationship of Cu, Zn, and Ca would not normally be consistent with the observed correlation. In particular, it would be expected for Cu, Zn, and S to correlate in sulfide mineral precipitates, but the ionic radius of Ca is large enough that it would not routinely be present within sulfides. Calcium would be expected to precipitate in carbonate minerals, which might include Zn and low quantities of Cu, but sulfur would be excluded. It is also clear from FTIR, VP-FEG-SEM, and X-ray diffraction (not shown) analyses that most of the sulfur imaged here is organic sulfur, and therefore sulfide mineral precipitation cannot explain these patterns. Crossed, branching detail can be resolved in all elements. Cu, Zn, and Ca may be derived from chelation by melanin or keratin, while sulfur is derived from beta-keratin itself which is extremely sulfur-rich (several weight percent).

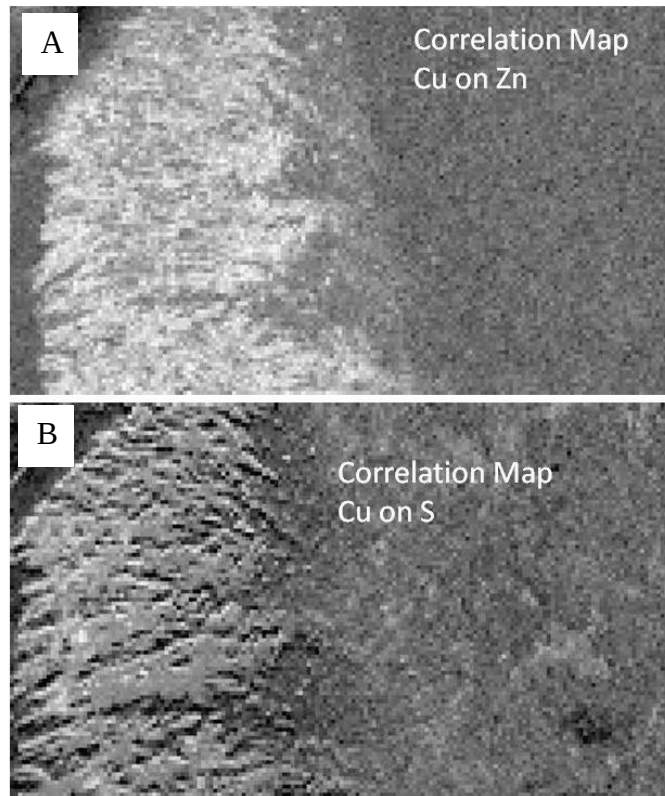


Fig. S3

Correlation maps for the *C. sanctus* neck feathers: A) Cu on Zn, B) Cu on S. Relatively strong correlation coefficients are calculated from the feather regions for Cu with Zn and S, such that maps of correlation coefficient strength (white: $R=1$, black: $R=-1$) clearly show the best correlations in the preserved feather regions, and indicate how elemental correlation is controlled by soft tissue structure. (Area shown is 3.5 mm x 2 mm.)

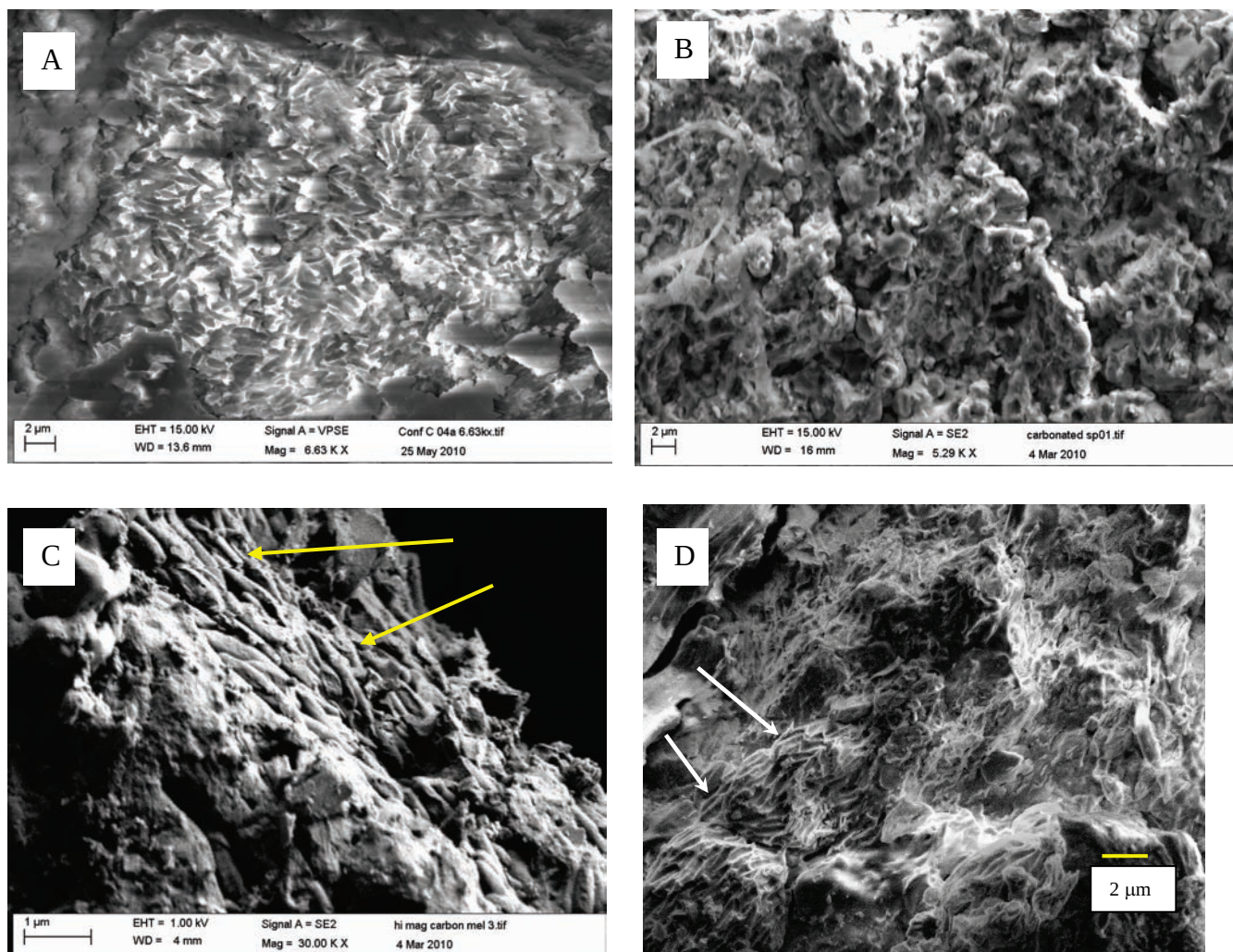


Fig. S4

VP-FEG-SEM images of fossils. A) Additional image of structures consistent with fossilized melanosomes correlated with high copper regions, here in the proximal wing feathers of *C. sanctus* (MGSF315). B) High copper region of Green River feather BHI-6358 with no discernible fossilized structures. Copper may provide a superior biomarker for pigmentation in comparison to SEM methods which seek to resolve structural details of preserved melanosomes. C) Images of Green River BHI-6319 feather: magnified view of preserved barb structure, fracture surface reveals aligned rod-like structures (arrows) inferred to be preserved melanosomes and present only in high-copper areas. D) High copper region of *G. yumenensis* feather (MGSF317) displaying potential melanosome structures (arrows). (SE detection mode at 3 kV accelerating voltage.)

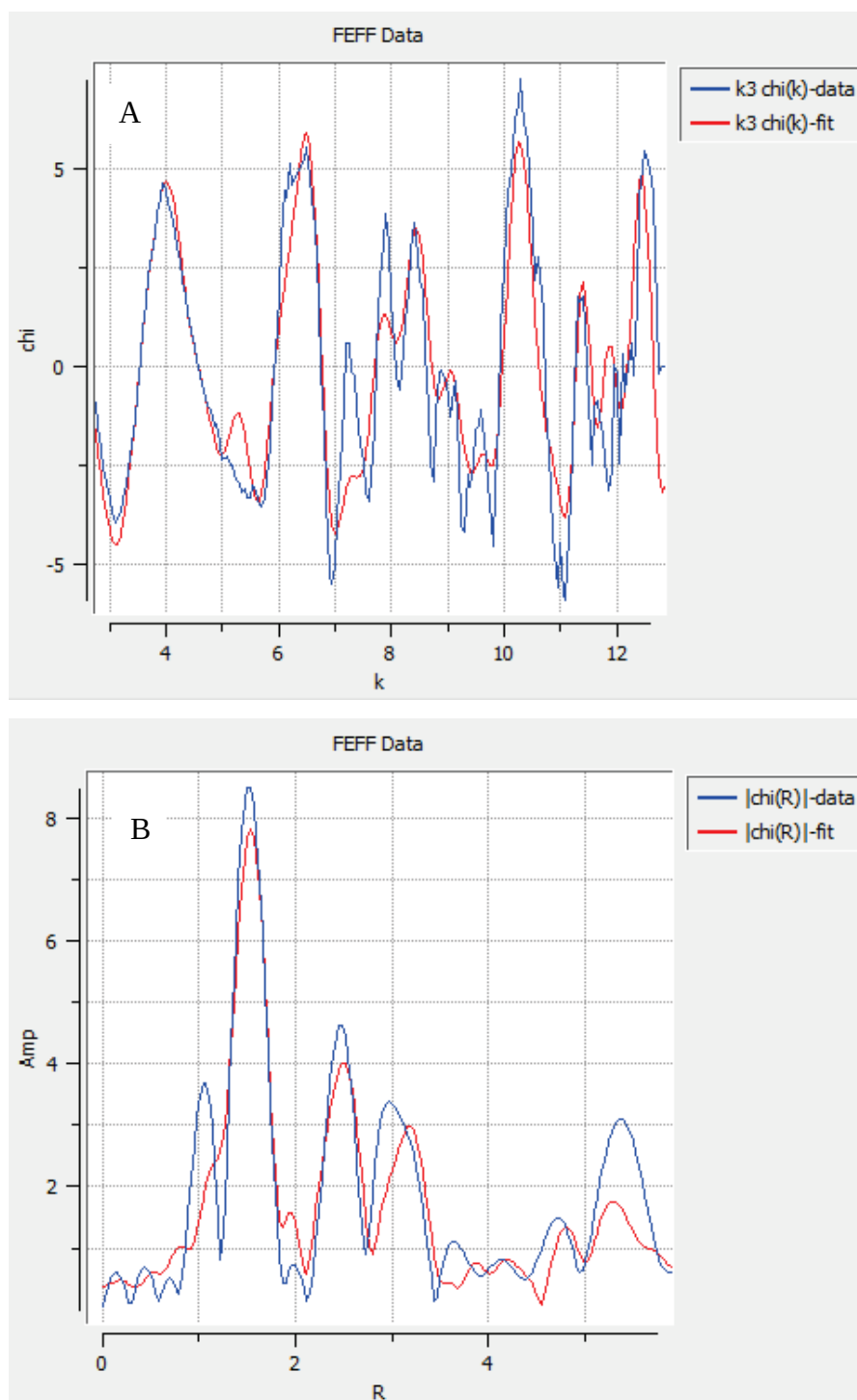


Fig. S5

A) EXAFS k-space data (blue line) and fit (red line) of the CuO standard corresponding to parameters given in the table above. B) Data (blue line) and fit (red line) for the CuO standard presented as a radial distribution.

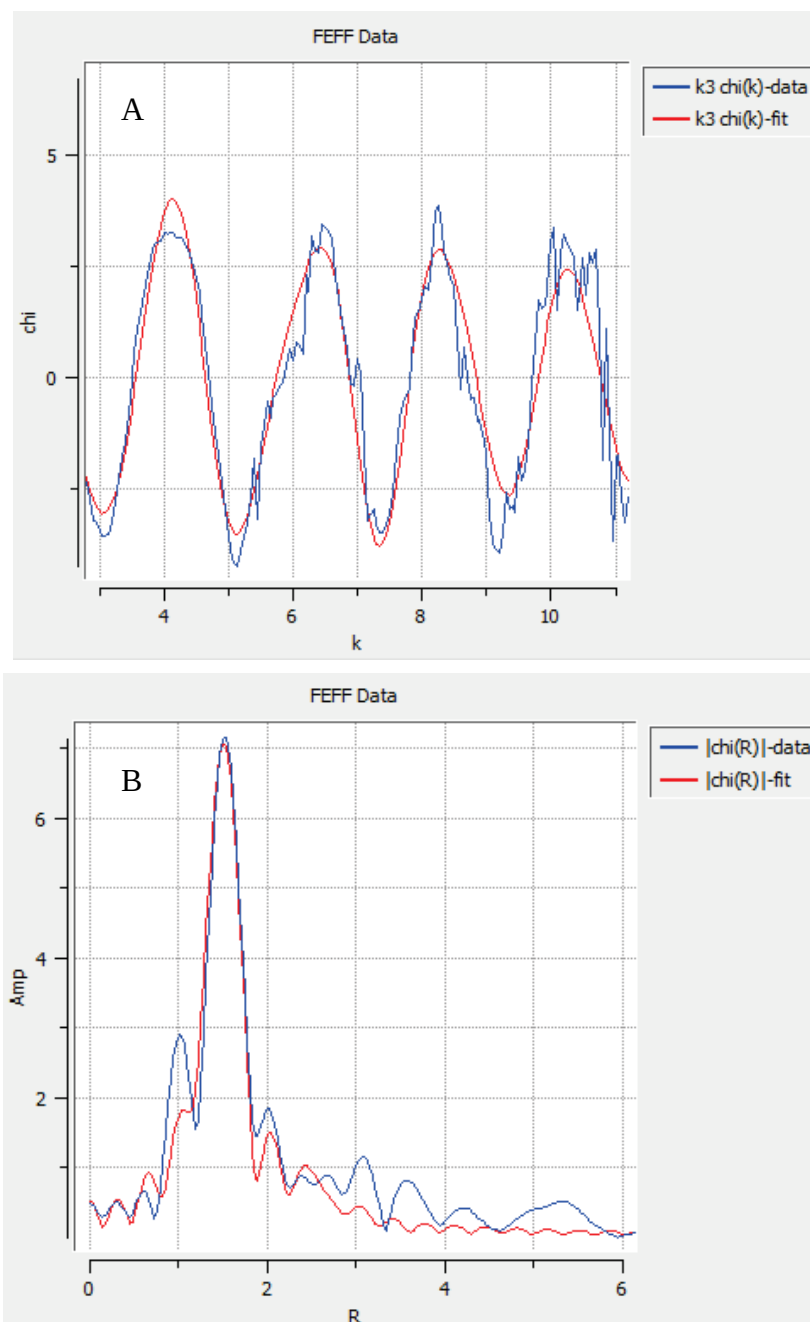


Fig. S6.

A) EXAFS k -space data (blue line) and fit (red line) of the Cu-Melanin standard corresponding to parameters given in the table above. B) Data (blue line) and fit (red line) for the Cu-Melanin standard presented as a radial distribution.

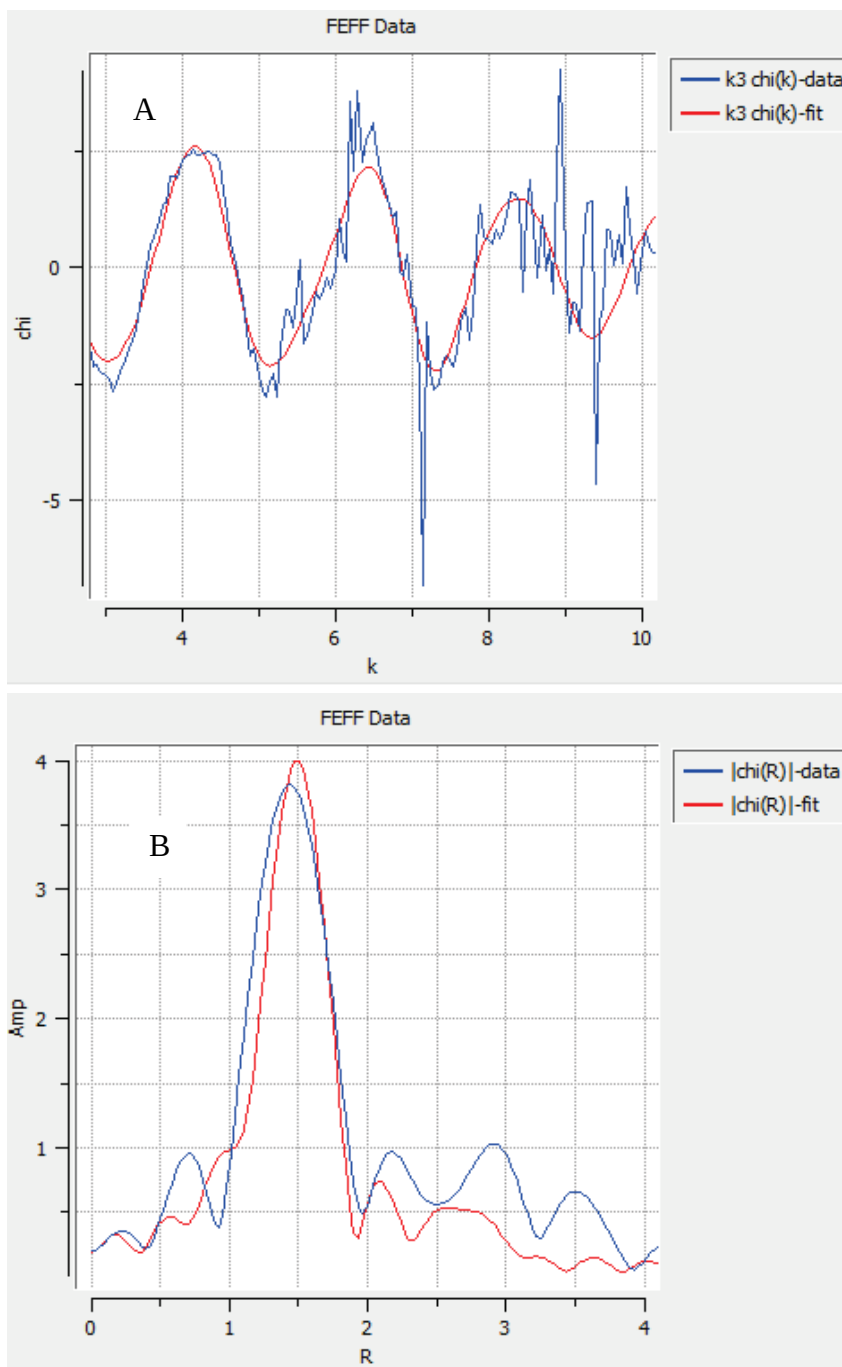


Fig. S7

A) EXAFS k-space data (blue line) and fit (red line) of copper in the *C. sanctus* specimen corresponding to parameters given in the table above. B) Data (blue line) and fit (red line) for the *C. sanctus* specimen presented as a radial distribution.

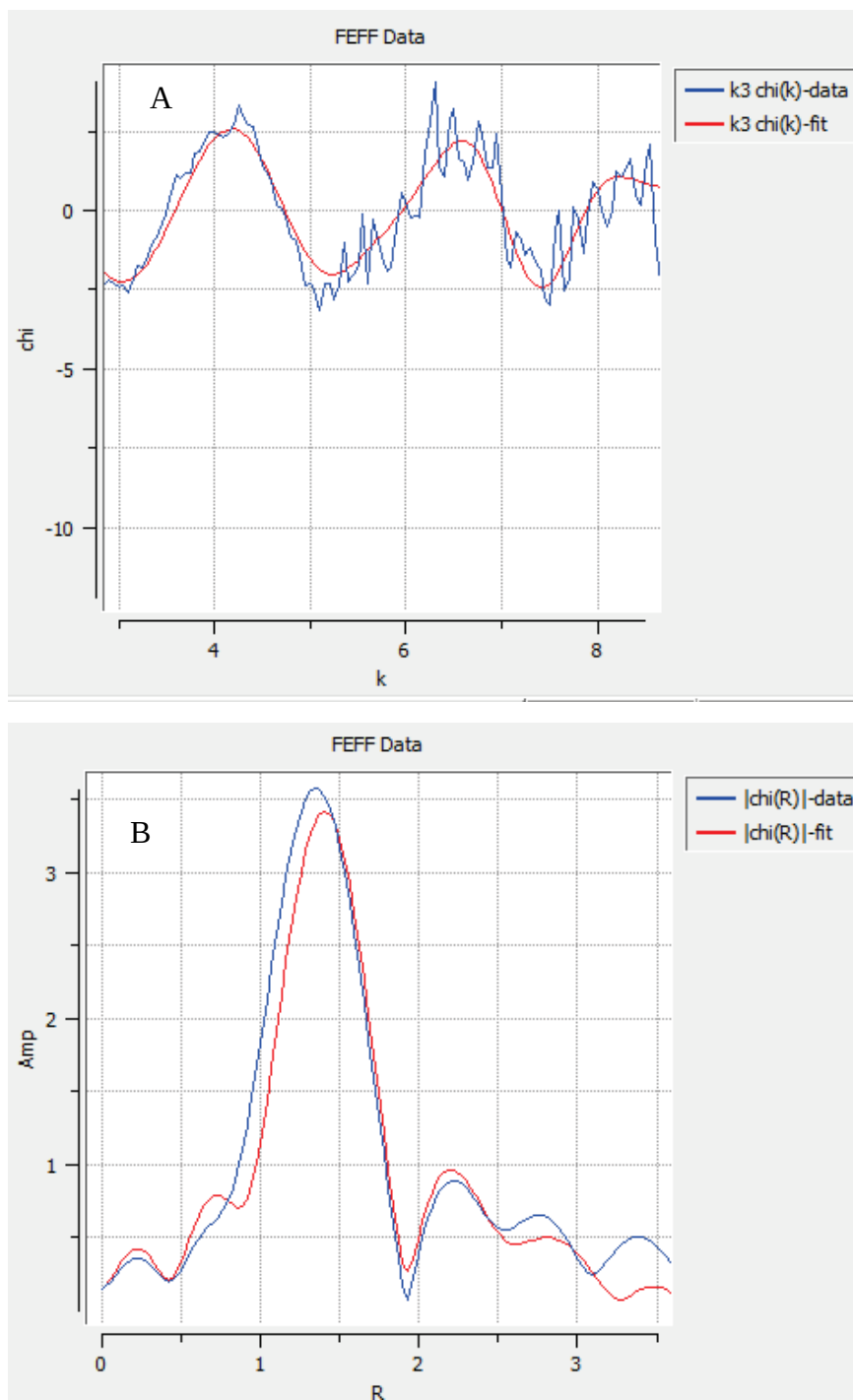


Fig. S8

A) EXAFS k -space data (blue line) and fit (red line) of copper in the Green River specimen (BHI-6358) corresponding to parameters given in the table above. B) Data (blue line) and fit (red line) for the Green River specimen presented as a radial distribution.

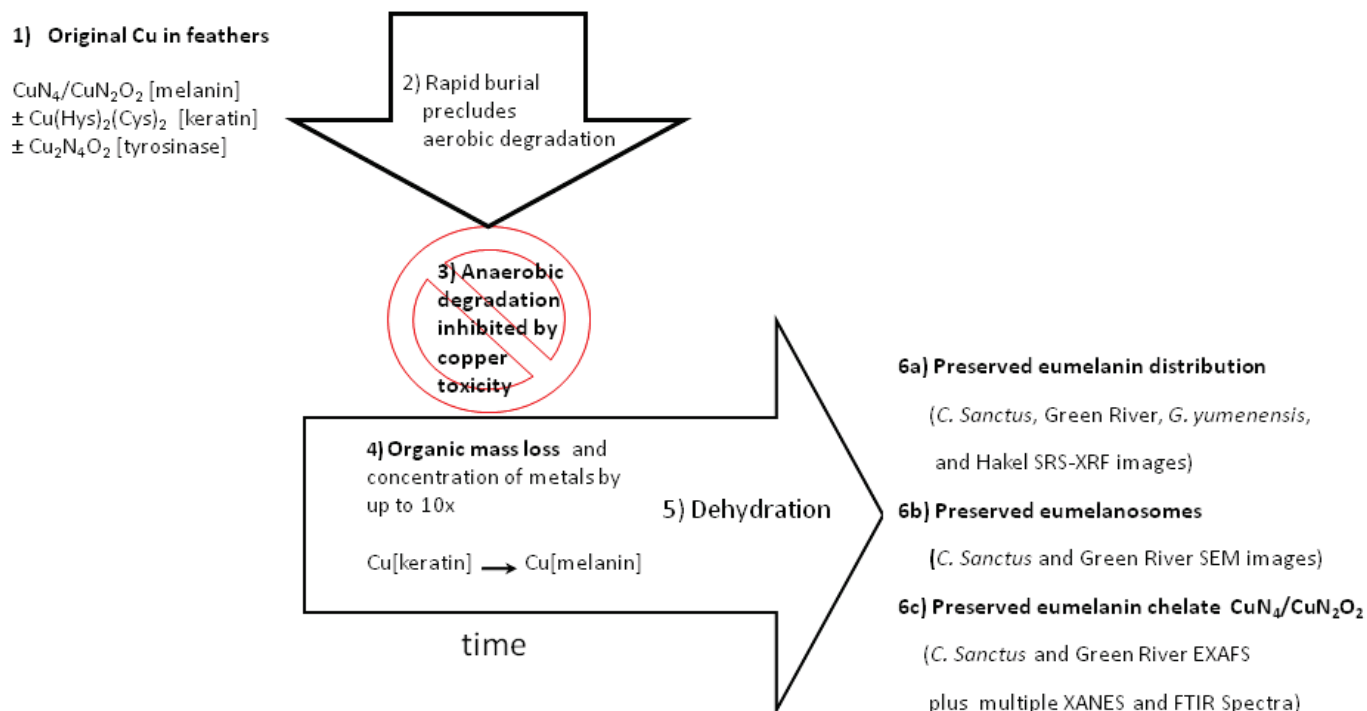


Fig. S9
 Taphonomic pathway for Cu-melanin chelates.

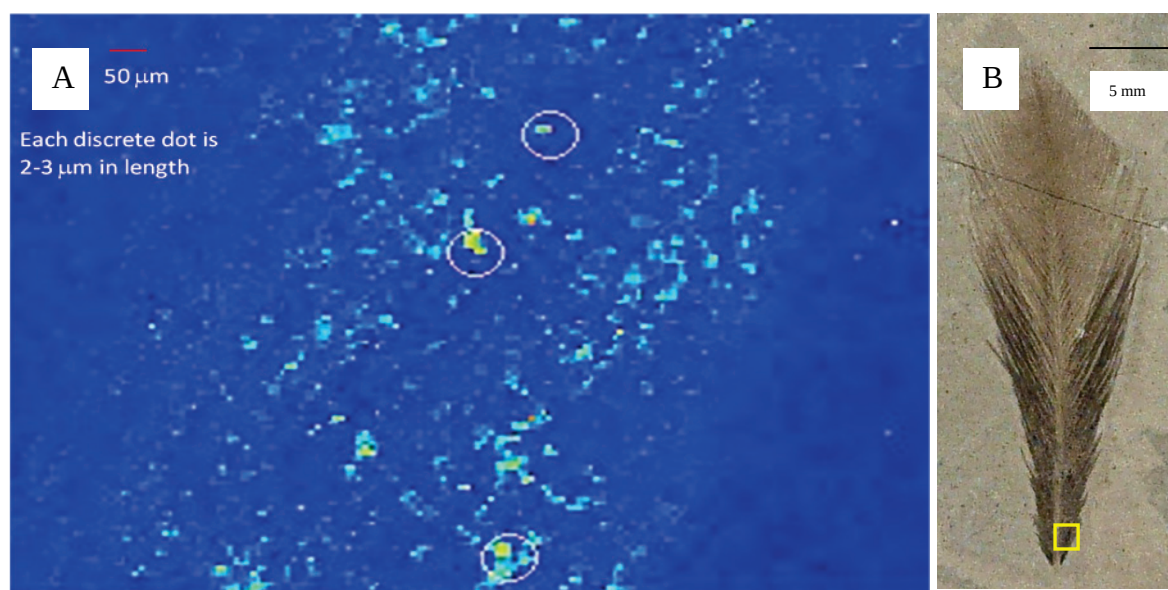


Fig. S10

A) Micro-XRF map of copper intensity in selected area of BHI-6358. Distribution of copper in patches a few microns in size may be due to discrete 2-3 μm size of the original melanosomes. B) Optical photograph of BHI-6358 showing approximate location of micro-XRF region (yellow box).

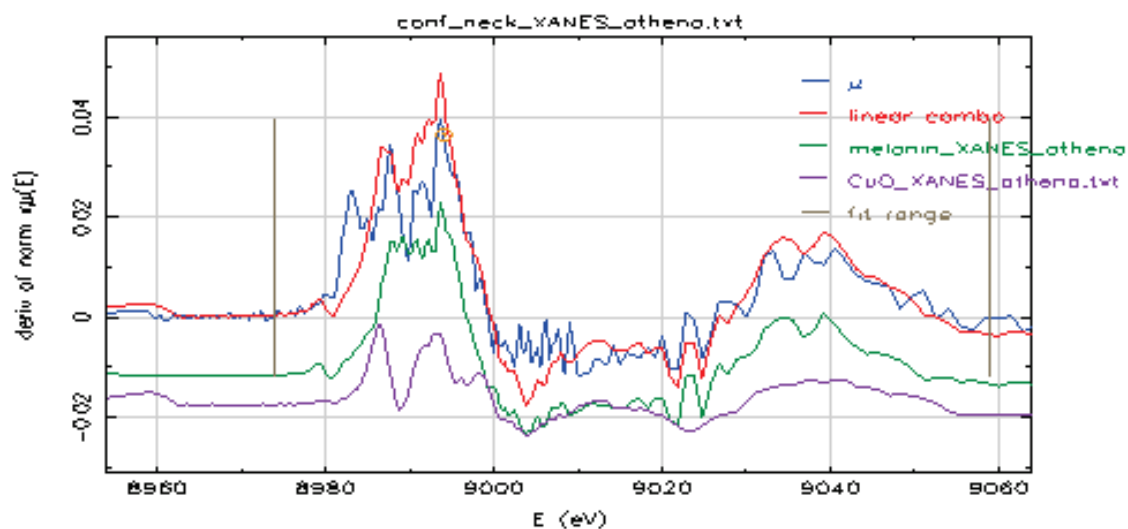
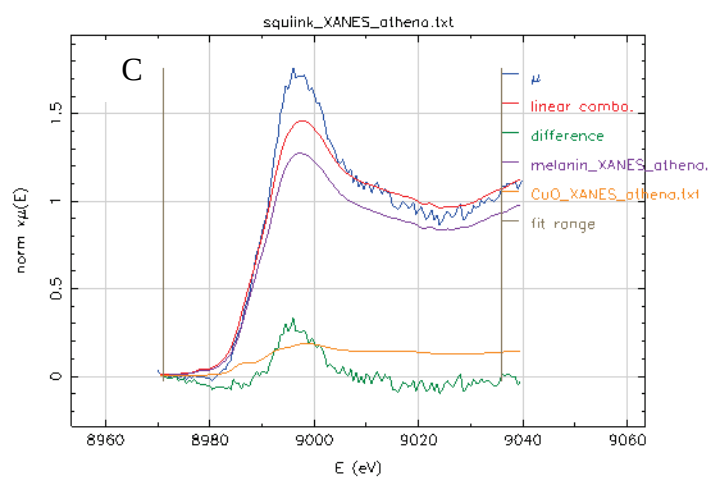
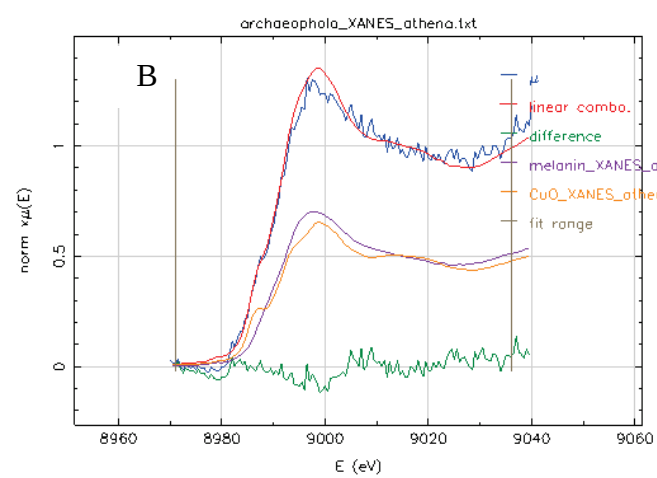
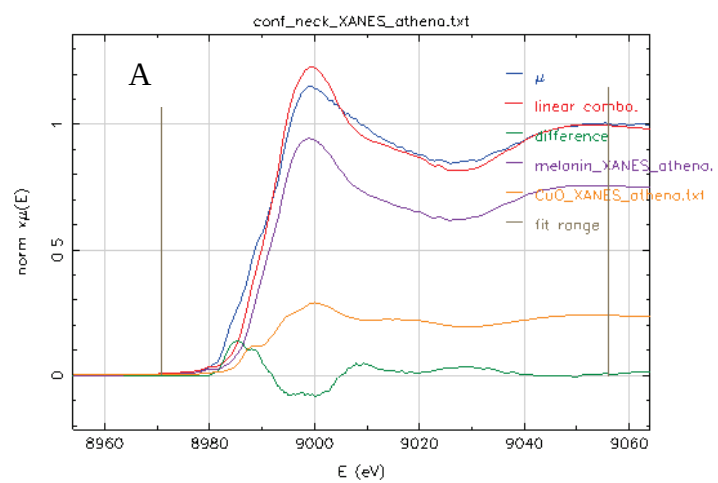


Fig. S11.

C. sanctus Linear Combination Analysis Fit in First Derivative Space. Fits were calculated using first derivative, but fit results are clearer when displayed as $\mu(E)$ and so the subsequent figures present the data and fits in that way.



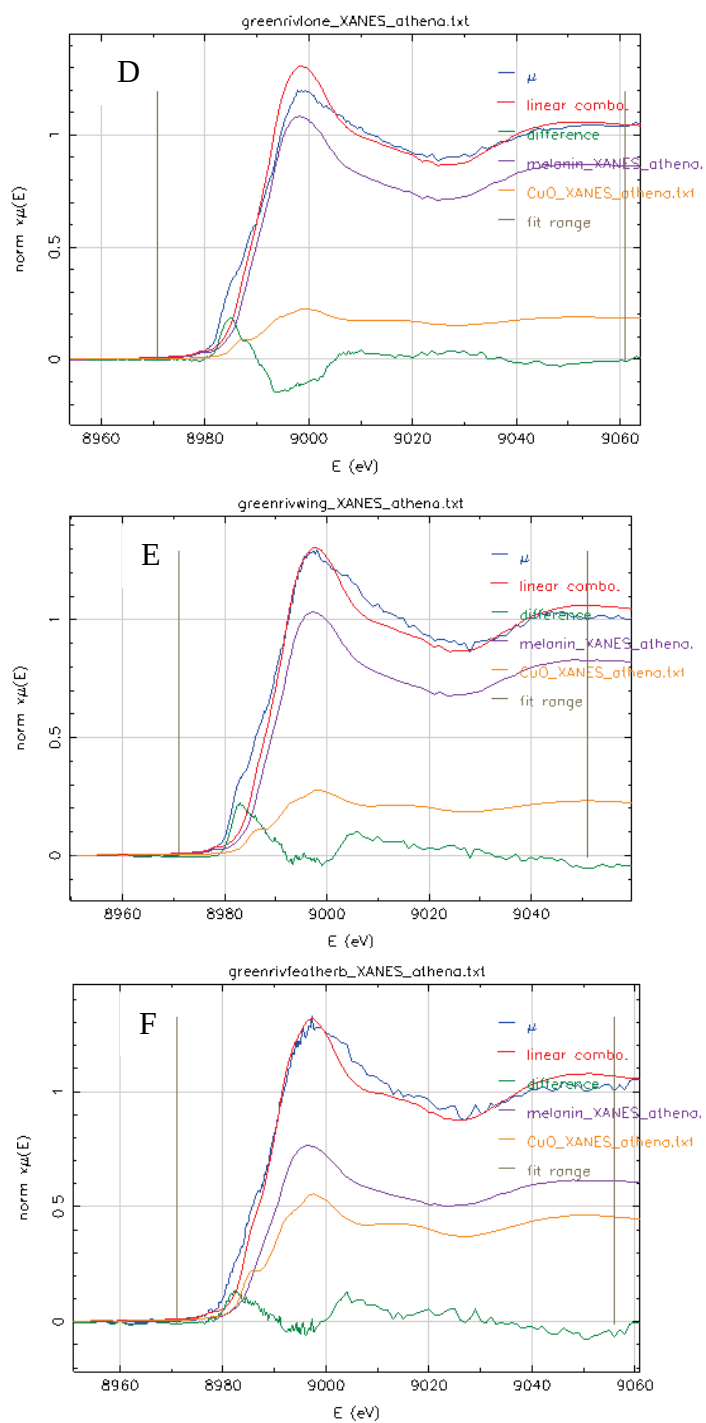


Fig. S12

XANES LCA results for fossils, all displayed as $\mu(E)$. A) *C. sanctus* neck, B) *Archaeopteryx* holotype, C) Fossil squid ink sack (BHI-2243B), D) Green River Feather (BHI-6358), E) Green River Feather (HMNS 2010.185.02), F) Green River Feather (BHI-6403). Blue=data, Red=LCA fit, Magenta=Cu-melanin component, Yellow=CuO component, Green= residuals.

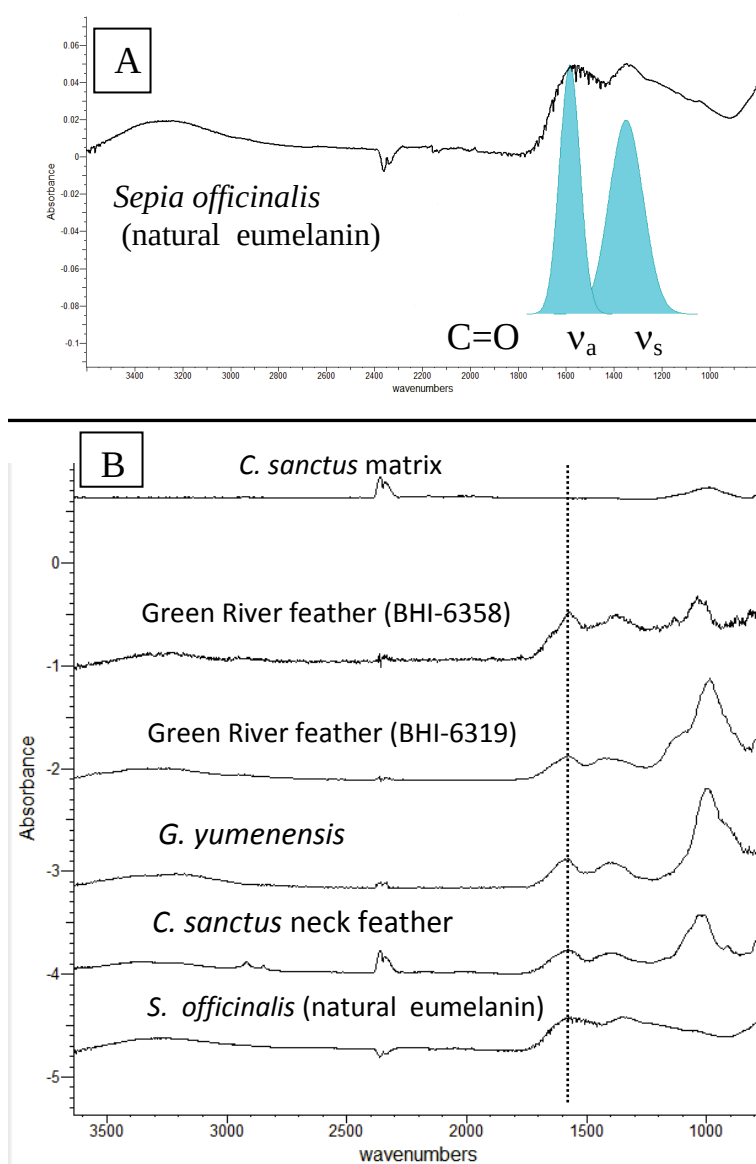


Fig. S13

A) ATR-FTIR spectrum of *Sepia officinalis*, a natural melanin standard. Carboxylate stretch absorption peak locations typical of eumelanin are labeled (ν_a , asymmetric; ν_s , symmetric) and indicated by the blue regions. C=C aromatic ring vibrations are convolved with the carboxylate absorption bands. See SOM Table S6 for full band assignments (37). Shoulder regions, especially to the high wavenumber side of the ν_a band may be caused by amide groups associated with protein derived impurities. B) ATR-FTIR spectra (normalized and offset) of the *C. sanctus* sedimentary matrix, four fossil feathers including *C. sanctus*, *G. yumenensis*, and two Green River feathers, compared to the *S. officinalis* eumelanin standard. The vertical dashed line shows the eumelanin asymmetric carboxylate stretch position at 1585 cm^{-1} for reference. Both of the absorption bands evident in the *S. officinalis* are present at nearly the same positions in all fossil feather samples. The *C. sanctus* matrix is representative of all of the matrices examined for these various samples in that it does not contain either of these bands. The broad intense peak present at approximately 1050 cm^{-1} in all of the fossils and in the *C. sanctus* matrix is due to silicate minerals, and the sharp peak present in some spectra at approximately 2360 cm^{-1} is atmospheric CO_2 . Interestingly, all of the fossils exhibit a shoulder to the high wavenumber side of the dominant organic peak, suggesting that amide groups derived from protein breakdown may also be present within the fossils.

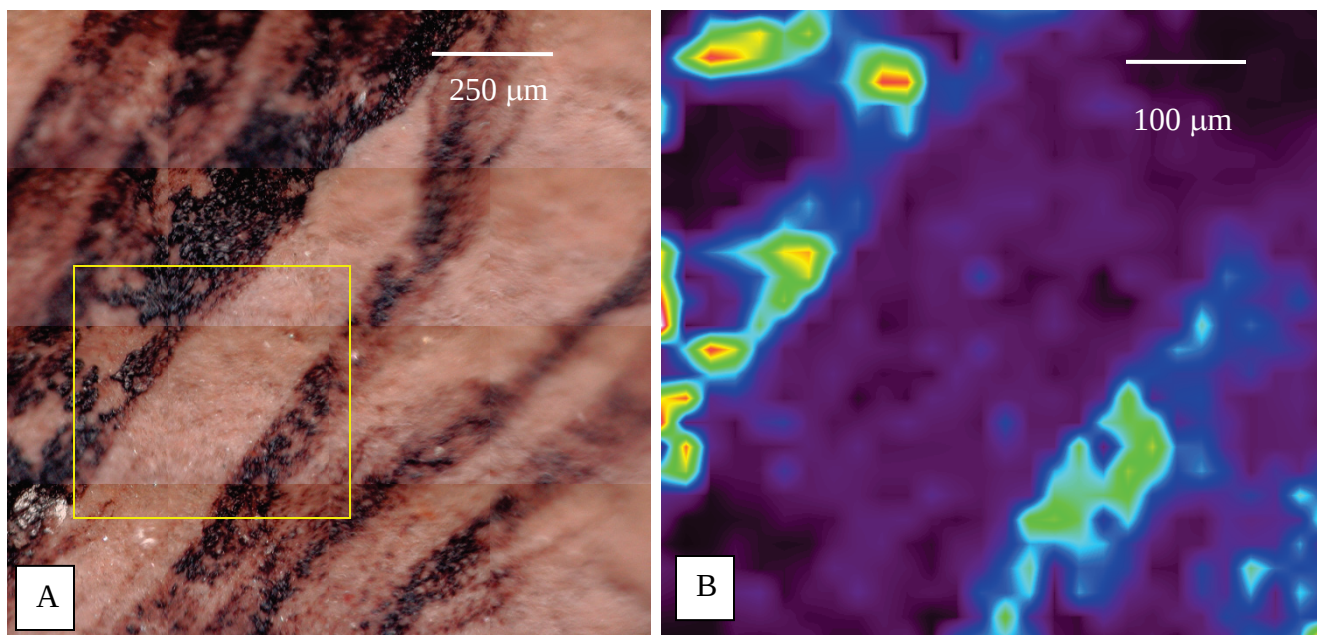


Fig. S14

A) Visible light image detail of Green River feather BHI-6319 showing preservational detail of barbs. B) ATR-FTIR map of the boxed area in (A) showing infra-red absorption at the wavenumber corresponding to the carboxylate asymmetric stretch (1585 cm^{-1}). This organic functional group is likely to be concentrated within the fossilized feather material.

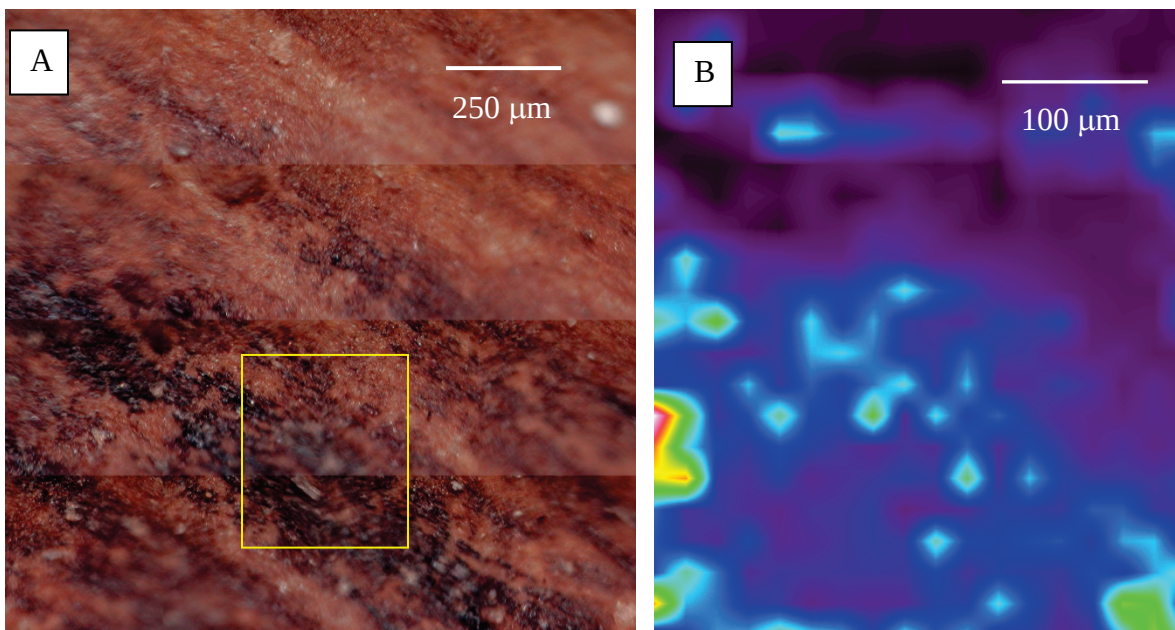


Fig. S15

A) Visible light image detail of *Gansus yumenensis* feather MGSF317 also showing preservational detail of barbs. B) ATR-FTIR map of the boxed area in (A) showing the infra-red absorption at the wavenumber corresponding to the carboxylate asymmetric stretch (1585 cm^{-1}). This organic functional group is again most likely concentrated within the fossilized feather material.

Table S1.

Correlation coefficients for image analysis of C. sanctus neck detail SRS-XRF maps

	<i>R</i>	<i>R</i> ²
Cu/Ca	0.75	0.563
Cu/S	0.69	0.476
Cu/Zn	0.61	0.372
Ca/Zn	0.24	0.058

Table S2.
EXAFS Fitting Details

Copper Oxide						
	n	R (Å)	σ^2	ΔE	χ^2_{ν}	So2
Cu-O (eq)	2.9	1.949	0.002			
Cu-O (ax)	3.2	2.879	0.001			
Cu-Cu	0.8	3.454	0.001			
Cu-Cu	6.2	5.282	0.009	-0.43	47.66	0.79
Natural Copper Melanin						
Cu-O/N (eq)	3.5	1.949	0.002			
Cu-O (ax)	2.0	2.783	0.009			
Cu-C	2.2	3.301	0.006	-1.25	9.27	0.64
<i>C. sanctus</i>, neck						
Cu-O/N (eq)	2.9	1.945	0.004			
Cu-O (ax)	1.2	2.835	0.009			
Cu-C	2.4	3.357	0.008	-0.71	3.83	0.54
Green River Feather (BHI-6358)						
Cu-O/N (eq)	3.3	1.915	0.005			
Cu-O (ax)	1.5	2.800	0.003			
Cu-C	2.3	3.257	0.003	-2.88	2.47	0.53

Table S3.

Cu, Zn, and C Levels in Fossils compared to Extant Organisms. Cu and Zn determined from SRS-XRF point analyses (ppm weight); C via VP-FEG-SEM point analysis (elemental %).

	Feather			"Original"	Matrix	
Fossils	Cu	Zn	C	Cu	Cu	Zn
<i>C. sanctus</i>	434	287	13.7-43.5%	130-400	16	67
Green River BHI-6358	152	115	14.1-46.4%	45-150	25	105
Green River BHI-6319	67	78	10.9-44.4%	15-65	49	90
<i>G. yumenesis</i>	138	96	1.9-42.7%	5-130	58	76
	Fish Eye					
Green River BHI-6319	89	108	10.9-42.1%	20-80		
Extant Feathers						
Eagle	0.6	749				
Blue Jay (maxima)	3	44				
Turaco (pigmented)	36,500	1506				
Turaco (white)	95	359				
White-Tailed Eagle [max.] (12)	16	260				
Other Extant Melanin						
<i>Sepia Officinalis</i> (38)	14	4				
Bovine Liver, acid treated melanin (38)	200	138				

Errors are approximately $\pm 20\%$ (relative) for metals, $\pm 4\%$ (absolute) for C.

Table S4.

Additional XRF and SEM Point Analyses (ppmW, except where % indicates element wt.%).

SRS-XRF	<i>Confuciusornis sanctus</i>			Green River					<i>G. yumenesis</i>			Eagle	Extant				
		(MGSF315)		BHI-6358		BHI-6319			(MGSF317)				Blue Jay			Turaco	
	Femur	Feather	Matrix	Feather	Matrix	Fish Eye	Matrix	Feather	Rachis	Barbs	Matrix		High Zn	Low Zn	Fe stripe	High Cu	Low Cu
Cu	4	434	16	152	25	89	49	67	138	103	58	0.6	3	2	2	36,500	95
Zn	82	287	67	115	105	108	90	78	96	82	76	749	44	10	31	1506	359
Fe	1458	1878	1029	1.81%	1.75%	3.32% ^a	3.32% ^a	3.32% ^a	4.74%	4.14%	3.60%	69	108	113	1002	1594	580
Ca	10.60%	3.57%	0.56%	4.33% ^a	4.33% ^a	7.68%	4.55%	4.00%	0.46%	0.58%	0.45%	0.12%	0.11%	0.10%	0.11%	1.97%	0.95%
Mn	424	22	bld	198	207	840	389	341	bld	21	bld	2	48	7	3	52	72
As	3	134	4	-	-	-	-	-	12	8	9	1	480	157	1217	1436	1160
S	16	2887	793	-	-	-	-	-	-	-	-	-	7.35%	-	7.90%	-	-
P	9.56%	1353	1398	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ca (SEM)		3.84%		4.16%		7.55%	5.14%	7.98%		6.75%	3.54%						
S (SEM)		5600				4500	900	8500		1.05%	0.10%						
^a Element used as internal standard																	
bld- below limit of detection																	
-' indicates no data																	
Errors are ±20% for metals determined by SRS-XRF. Errors on S and P are approximately a factor of 2.																	
Errors on Ca and S determined by SEM are approximately ±2 wt % for Ca and ±2000 ppmW for S.																	

Table S5.Linear Combination Analysis of Fossil Cu-XANES (proportion error ± 0.10)

	Cu- Melanin	CuO	χ^2
Fossil Squid Ink Sac (BHI-2243B)	0.87	0.13	0.05
Green River Feather (BHI-6358)	0.83	0.17	0.01
Green River Feather (HMNS 2010.185.02)	0.78	0.22	0.01
Green River Feather (BHI-6403)	0.57	0.43	0.02
<i>C. sanctus</i> neck	0.77	0.23	0.004
<i>Archaeopteryx</i> Holotype	0.51	0.49	0.05

Table S6.
Infra-red Absorption Band Reference Values (cm^{-1})

Melanin Assignments*

NH stretch 3375

Carboxylate

C=O antisymmetric stretch 1600-1560

C=O symmetric stretch 1420-1400

Aromatic Ring C=C 1642, 1614, 1600, 1560, 1545

CH in-plane deformation 1074

[*Only very strong and strong bands (37) listed]

Other possible groups

Si-O-Si stretch 1100-1000

Organic Sulfur, SO_2 antisymmetric stretch 1415-1390

SO_2 symmetric stretch 1200-1187

S-O antisymmetric stretch 1020-850

CO_2 2380-2350

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